

Chapter 20

Underdetermined Systems

*I'm thinking of two numbers.
Their average is 3.
What are the numbers?*

— CLEVE B. MOLER, *The World's Simplest Impossible Problem* (1990)

*This problem arises in important algorithms
used in mathematical programming . . .
In these cases, B is usually very large and sparse and,
because of storage difficulties,
it is often uneconomical to store and access Q_1 . . .
Sometimes it has been thought that [the seminormal equations method]
could be disastrously worse than [the Q method] . . .
It is the purpose of this note to show that such algorithms are
numerically quite satisfactory.*

— C. C. PAIGE, *An Error Analysis of a
Method for Solving Matrix Equations* (1973)

Having considered well-determined and overdetermined linear systems, we now turn to the remaining class of linear systems: those that are underdetermined.

20.1. Solution Methods

Consider the underdetermined system $Ax = b$, where $A \in \mathbf{R}^{m \times n}$ with $m \leq n$. The system can be analysed using a QR factorization

$$A^T = Q \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad (20.1)$$

where $Q \in \mathbf{R}^{n \times n}$ is orthogonal and $R \in \mathbf{R}^{m \times m}$ is upper triangular. (We could, alternatively, use an LQ factorization of A , but we will keep to the standard notation.) We have

$$b = Ax = [R^T \ 0]Q^T x = R^T y_1, \quad (20.2)$$

where

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = Q^T x, \quad y_1 \in \mathbf{R}^m.$$

If A has full rank then $y_1 = R^{-T}b$ is uniquely determined and all solutions of $Ax = b$ are given by

$$x = Q \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad y_2 \in \mathbf{R}^{n-m} \text{ arbitrary.}$$

The unique solution x_{LS} that minimizes $\|x\|_2$ is obtained by setting $y_2 = 0$. We have

$$x_{LS} = Q \begin{bmatrix} R^{-T}b \\ 0 \end{bmatrix} \quad (20.3)$$

$$= Q \begin{bmatrix} R \\ 0 \end{bmatrix} R^{-1} R^{-T}b = Q \begin{bmatrix} R \\ 0 \end{bmatrix} (R^T R)^{-1}b \quad (20.4)$$

$$\begin{aligned} &= A^T (AA^T)^{-1}b \\ &= A^+ b, \end{aligned}$$

where $A^+ = A^T (AA^T)^{-1}$ is the pseudo-inverse of A . Hence x_{LS} can be characterized as $x_{LS} = A^T y$, where y solves the *normal equations* $AA^T y = b$.

Equation (20.3) defines one way to compute x_{LS} . We will refer to this method as the “Q method”. When A is large and sparse it is desirable to avoid storing and accessing Q , which can be expensive. An alternative method

with this property uses the QR factorization (20.1) but computes x_{LS} as $x_{LS} = A^T y$, where

$$R^T R y = b \quad (20.5)$$

(cf. (20.4)). These latter equations are called the *seminormal equations* (SNE). As the “semi” denotes, however, this method does not explicitly form AA^T , which would be undesirable from the standpoint of numerical stability. Note that equations (20.5) are different from the equations $R^T R x = A^T b$ for an overdetermined least squares (LS) problem, where $A = Q[R^T \ 0]^T \in \mathbb{R}^{m \times n}$ with $m \geq n$, which are also called seminormal equations (see §19.6).

20.2. Perturbation Theory

A componentwise perturbation result for the minimum 2-norm solution to an underdetermined system is readily obtained.

Theorem 20.1 (Demmel and Higham). *Let $A \in \mathbb{R}^{m \times n}$ ($m \leq n$) be of full rank and $0 \neq b \in \mathbb{R}^m$. Suppose $\|A^+ \Delta A\|_2 < 1$ and*

$$|\Delta A| \leq \epsilon E, \quad |\Delta b| \leq \epsilon f.$$

If x and y are the minimum 2-norm solutions to $Ax = b$ and $(A + \Delta A)y = b + \Delta b$, respectively, then, for any monotonic norm,

$$\frac{\|x - y\|}{\|x\|} \leq \left(\|I - A^+ A\| \cdot E^T \cdot \|A^{+T} x\| + \|A^+\| \cdot (f + E\|x\|) \right) \frac{\epsilon}{\|x\|} + O(\epsilon^2). \quad (20.6)$$

For any Hölder p -norm, the bound is attainable to within a constant factor depending on n .

Proof. The perturbed matrix $A + \Delta A = A(I + A^+ \Delta A)$ has full rank, so we can manipulate the equation

$$y = (A + \Delta A)^T ((A + \Delta A)(A + \Delta A)^T)^{-1} (b + \Delta b)$$

to obtain

$$\begin{aligned} y - x &= (I - A^+ A) \Delta A^T (A A^T)^{-1} b + A^+ (\Delta b - \Delta A x) + O(\epsilon^2) \\ &= (I - A^+ A) \Delta A^T A^{+T} x + A^+ (\Delta b - \Delta A x) + O(\epsilon^2). \end{aligned} \quad (20.7)$$

The required bound follows on using absolute value inequalities and taking norms. That the bound is attained to within a constant factor depending on n for Hölder p -norms is a consequence of the fact that the two vectors on the right-hand side of (20.7) are orthogonal. $\square$

Two special cases are worth noting, for later use. We will use the equality $\|I - A^+A\|_2 = \min\{1, n - m\}$, which can be derived by consideration of the QR factorization (20.1), for example. If $E = |A|H$, where H is a given nonnegative matrix, and $f = |b|$, then we can put (20.6) in the form

$$\frac{\|x - y\|_2}{\|x\|_2} \leq \min\{3, n - m + 2\} \max\{\|H\|_2, 1\} \text{cond}_2(A)\epsilon + O(\epsilon^2), \quad (20.8)$$

where

$$\text{cond}_2(A) = \| |A^+| |A| \|_2.$$

Note that $\text{cond}_2(A)$ is independent of the row scaling of A ($\text{cond}_2(DA) = \text{cond}_2(A)$ for nonsingular diagonal D). If $E = \|A\|_2 e_m e_n^T$ and $f = \|b\|_2 e_m$, where e_m denotes the m -dimensional vector of 1s, then

$$\frac{\|x - y\|_2}{\|x\|_2} \leq \min\{3, n - m + 2\} (mn)^{1/2} \kappa_2(A) \epsilon + O(\epsilon^2). \quad (20.9)$$

The following analogue of Lemma 19.6 will be needed for the error analysis in the next section. It says that if we perturb the two occurrences of A in the normal equations $AA^T x = b$ differently, then the solution of the perturbed system is the solution of normal equations in which there is only one perturbation of A and, moreover, this single perturbation is no larger, in the normwise or componentwise sense, than the two perturbations we started with.

Lemma 20.2 (Kielbasiński and Schwetlick). *Let $A \in \mathbf{R}^{m \times n}$ ($m \leq n$) be of full rank and suppose*

$$(A + \Delta A_1) \bar{x} = b, \quad \bar{x} = (A + \Delta A_2)^T \tilde{y}.$$

Assume that $3 \max(\|A^+ \Delta A_1\|_2, \|A^+ \Delta A_2\|_2) < 1$. Then there is a vector $\tilde{y}$ and a perturbation ΔA with

$$\Delta A = \Delta A_1 G_1 + \Delta A_2 G_2, \quad G_1^T = G_1 = G_2^T, \quad G_1 + G_2 = I,$$

such that

$$(A + \Delta A) \bar{x} = b, \quad \bar{x} = (A + \Delta A)^T \tilde{y},$$

that is, $\bar{x}$ is the minimum 2-norm solution to $(A + \Delta A)x = b$.

Proof. The proof is similar to that of Lemma 19.6, but differs in some details. If $\bar{x} = (A + \Delta A_2)^T \tilde{y} = 0$ we take $\Delta A = \Delta A_2$. otherwise, we set

$$\Delta A := \Delta A_1 P + \Delta A_2 (I - P) =: \Delta A_2 + HP,$$

where $P \doteq \bar{x}\bar{x}^T/\bar{x}^T\bar{x}$ and $H = \Delta A_1 - \Delta A_2$. We have

$$(A + \Delta A)^T \tilde{y} = (A + \Delta A_2 + HP)^T \tilde{y} = \beta \bar{x},$$

where $\beta = 1 + \bar{x}^T H^T \tilde{y} / \bar{x}^T \bar{x}$, which shows that we need to set $\tilde{y} = \beta^{-1} \bar{y}$. To check that $(A + \Delta A)\bar{x} = b$, we evaluate

$$\begin{aligned} (A + \Delta A)\bar{x} &= (A + \Delta A_1 + H(P - I))\bar{x} \\ &= (A + \Delta A_1)\bar{x} = b, \end{aligned}$$

as required. The vector $\tilde{y}$ is undefined if $\beta = 0$. But

$$\begin{aligned} \beta &= 1 + \frac{\bar{x}^T H^T \tilde{y}}{\bar{x}^T \bar{x}} = 1 + \frac{\bar{x}^T H^T (A + \Delta A_2)^+{}^T \bar{x}}{\bar{x}^T \bar{x}} \\ &\geq 1 - \|(A + \Delta A_2)^+ H\|_2 \\ &\geq 1 - \frac{\|A^+ H\|_2}{1 - \|A^+ \Delta A_2\|_2} \quad (\text{if } \|A^+ \Delta A_2\|_2 < 1) \\ &\geq 1 - \frac{(\|A^+ \Delta A_1\|_2 + \|A^+ \Delta A_2\|_2)}{1 - \|A^+ \Delta A_2\|_2}, \end{aligned}$$

which is positive if $3 \max(\|A^+ \Delta A_1\|_2, \|A^+ \Delta A_2\|_2) < 1$. $\square$

Note that in Lemma 20.2 we have the normwise bound

$$\|\Delta A\|_p \leq (\|\Delta A_1\|_p^2 + \|\Delta A_2\|_p^2)^{1/2}, \quad p = 2, F.$$

20.3. Error Analysis

We now consider the stability of the Q method and the SNE method. For both methods we assume that the QR factorization is computed using Householder or Givens transformations.

Before presenting the results we define a measure of stability. The componentwise backward error for a minimum-norm underdetermined system $Ax = b$ is defined as

$$\begin{aligned} \bar{\omega}_{E,f}(y) &:= \min\{\epsilon : \exists \Delta A \in \mathbb{R}^{m \times n}, \Delta b \in \mathbb{R}^m \text{ with } |\Delta A| \leq \epsilon E, |\Delta b| \leq \epsilon f \\ &\quad \text{s.t. } y \text{ is the min. norm solution to } (A + \Delta A)y = b + \Delta b\}, \end{aligned}$$

Note the requirement in this definition that y be the minimum norm solution; the usual componentwise backward error $\omega_{E,f}(y)$ (see (7.6)) is a generally smaller quantity. Let us say that a method is row-wise *backward stable* if it produces a computed solution $\hat{x}$ for which the componentwise backward error

$\bar{\omega}_{E,f}$ is of order u , where $E = |A|e_me_m^T$ and $f = |b|$. This condition requires that $\hat{x}$ solve a perturbed minimum norm problem in which the perturbations to the i th row of A are small compared with the norm of that row (similarly for b); cf. the discussion of componentwise backward errors in §7.2.

Theorem 20.3. *Let $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m \leq n$, and assume that a rendition of the form $\text{cond}_2(A)mn\gamma_{cn} < 1$ holds. Suppose the underdetermined system $AX = b$ is solved in the minimum 2-norm sense using the Q method. Then the computed solution $\hat{x}$ is the minimum 2-norm solution to $(A + \Delta A)x = b$, where*

$$\|\Delta A\|_F \leq m\gamma_{cn}\|A\|_F$$

and

$$|\Delta A| \leq mn\gamma_{cn}|A|G, \quad \|G\|_F = 1.$$

Proof. The Q method solves the triangular system $R^T y_l = b$ and then forms $x = Q[y_1^T, 0]^T$. Assuming the use of Householder QR factorization, from Theorem 18.4 we have that

$$(A + \Delta A_0)^T = Q \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix},$$

for some orthogonal matrix Q, where $\|\Delta A_0\|_F \leq m\gamma_{cn}\|A\|_F$ and $|\Delta A_0^T| \leq mn\gamma_{cn}G_0^T|A^T|$ with $\|G_0\|_F = 1$. The computed $\hat{y}_1$ satisfies

$$(\hat{R} + \Delta \hat{R})^T \hat{y}_1 = b, \quad |\Delta \hat{R}| \leq \gamma_m |\hat{R}|.$$

From Lemma 18.3, the computed solution $\hat{x}$ satisfies

$$\hat{x} = (Q + \Delta Q) \begin{bmatrix} \hat{y}_1 \\ 0 \end{bmatrix}, \quad \|\Delta Q\|_F \leq m\gamma_{cn}. \quad (20.10)$$

We now rewrite the latter two equations in such a way that we can apply Lemma 20.2:

$$\begin{aligned} (A + \Delta A_1)\hat{x} &:= [(\hat{R} + \Delta \hat{R})^T \ 0] (Q + \Delta Q)^{-1} \cdot (Q + \Delta Q) \begin{bmatrix} \hat{y}_1 \\ 0 \end{bmatrix} = b, \\ \hat{x} &= (Q + \Delta Q) \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} \cdot \hat{R}^{-1} \hat{y}_1 =: (A + \Delta A_2)^T \bar{y}. \end{aligned}$$

It is straightforward to show that

$$|\Delta A_i| \leq m\gamma'_{cn}|A|G_i, \quad \|G_i\|_F = 1, \quad i = 1:2,$$

and $\|\Delta A_i\|_F \leq m\gamma'_i\|A\|_F$, $i = 1:2$. The result follows on invocation of Lemma 20.2. $\square$

Theorem 20.3 says that the Q method is row-wise backward stable. This is not altogether surprising, since (Householder or Givens) QR factorization for the LS problem enjoys an analogous backward stability result (Theorem 19.3), albeit without the restriction of a minimum norm solution. Applying (20.8) to Theorem 20.3 we obtain the forward error bound

$$\frac{\|\hat{x} - x\|_2}{\|x\|_2} \leq mn\gamma'_{cn} \text{cond}_2(A) + O(u^2). \quad (20.11)$$

The same form of forward error bound (20.11) can be derived for the SNE method as for the Q method [292, 1993]. However, it is not possible to obtain a result analogous to Theorem 20.3, nor even to obtain a residual bound of the form $\|b - A\hat{x}\|_2 \leq c_{m,n}u\|A\|_2\|\hat{x}\|_2$ (which would imply that $\hat{x}$ solved a nearby system, though $\hat{x}$ would not necessarily be the minimum norm solution). The method of solution guarantees only that the seminormal equations themselves have a small residual. Thus, as in the context of overdetermined LS problems, the SNE method is not backward stable. A possible way to improve the stability is by iterative refinement, as shown in [292, 1993].

Note that the forward error bound (20.11) is independent of the row scaling of A , since $\text{cond}_2(A)$ is. The bound is therefore potentially much smaller than the bound

$$\frac{\|x - \hat{x}\|_2}{\|x\|_2} \leq c_{m,n}u\kappa_2(A) + O(u^2),$$

obtained by Paige [813, 1973] for the SNE method and by Jennings and Osborne [614, 1974] and Arioli and Laratta [26, 1985, Thin. 4] for the Q method.

Finally, we mention an alternative version of the Q method that is based on the modified Gram–Schmidt (MGS) method. The obvious approach is to compute the QR factorization $AT = QR$ using MGS ($Q \in \mathbb{R}^{n \times m}$, $R \in \mathbb{R}^{m \times m}$) solve $R^T y = b$, and then form $x = Qy$. Since Q is provided explicitly by the MGS method, the final stage is a full matrix–vector multiplication, unlike for the Householder method. However, because the computed Q may depart from orthonormality, this method is unstable in the form described. The formation of $x = Qy$ should instead be done as follows:

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 $x^{(n)} = 0$ 
for  $k = n: -1: 1$ 
     $\alpha_k = q_k^T x^{(k)}$ 
     $x^{(k-1)} = x^{(k)} - (\alpha_k - y_k)q_k$ 
end
 $x = x^{(0)}$ 

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The recurrence can be written as $x^{(k-1)} = x^{(k)} + y_k q_k - (q_k^T x^{(k)})q_k$, and the last term is zero in exact arithmetic if the q_k are mutually orthogonal. In finite precision arithmetic this correction term has the “magical” effect of making

Table 20.1. *Backward errors for underdetermined Vandermonde system.*

	$\eta_{A,b}(\hat{x})$
Householder QR	9.76×10^{-18}
MGS with $x := Qy$	4.10×10^{-4}
MGS with x formed stably (see text)	2.25×10^{-17}
SNE method (using Householder QR)	1.99×10^{-4}

the algorithm stable, in the sense that it satisfies essentially the same result as the Q method in Theorem 20.3; see Björck and Paige [120, 1994].

A numerical example is instructive. Take the 20×30 Vandermonde matrix $A = (p_i^{j-1})$, where the p_i are equally spaced on $[0, 1]$, and let b have elements equally spaced on $[0, 1]$. The condition number $\kappa_2(A) = 4.35 \times 10^{14}$. The (standard) normwise backward errors in the 2-norm are shown in Table 20.1. For A^T , the $\hat{Q}$ supplied by MGS satisfies $\|\hat{Q}^T \hat{Q} - I\|_2 = 1.41 \times 10^{-3}$, which explains the instability of the “obvious” MGS solver.

20.4. Notes and References

The seminormal equations method was suggested by Gill and Murray [442, 1973] and Saunders [894, 1972]. Other methods for obtaining minimal 2-norm solutions of underdetermined systems are surveyed by Cline and Plemmons [219, 1976].

Theorem 20.1 is from Demmel and Higham [292, 1993]. The bound (20.9) is well known; it follows from Wedin’s original version of our Theorem 19.1, which applies to minimum 2-norm underdetermined problems as well as LS problems.

Theorem 20.3 is new. Demmel and Higham [292, 1993] prove the weaker result that $\hat{x}$ from the Q method is *very close to* a vector $\bar{x}$ that satisfies the criterion for row-wise backward stability, and Lawson and Hanson [695, 1974, Thin. 16. 18] give a corresponding result in which $\bar{x}$ satisfies the criterion for general normwise backward stability. The key to showing actual backward stability is the use of Kielbasinski and Schwetlick’s lemma, which is a modification of Lemma 8.2.11 in [658, 1988] and [659, 1992] (our Lemma 19.6). Demmel and Higham [292, 1993] also give error analysis for the seminormal equations method.

The new MGS algorithm for solving the minimum norm problem was first suggested by Björck and Paige [119, 1992]; see also Björck [115, 1994].

Arioli and Laratta [27, 1986] give error analysis of QR factorization methods for solving the general problem $\min\{\|x - c\|_2 : A_x = b\}$, where $A \in \mathbb{R}^{m \times n}$